

Cell paths before layers

Lee maze finds a shortest path on grid cells ignoring track congestion

The idea

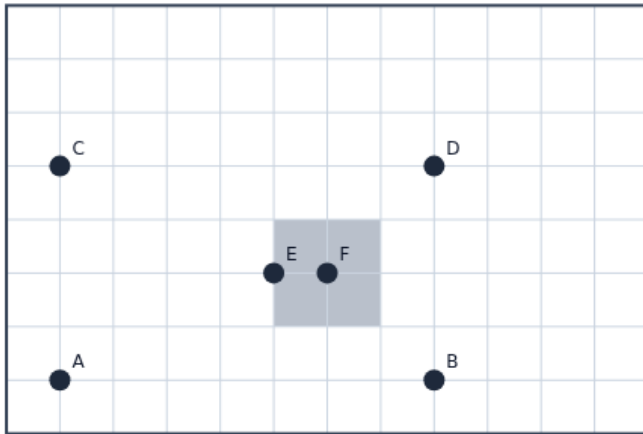
- Seed a BFS queue with start and path list
- Pop, expand neighbors4 skipping blocked cells, return the first path that reaches the goal
- If the queue empties, return None
- Shortest cell hop count wins

Breadth-first

LEE MAZE

Breadth-first

Step 1 / 5 · bfs · module02-01-lee-maze



Explanation

Lee expands uniformly—first arrival is shortest hop count.

- Grid graph
- Blockage aware

BFS layers

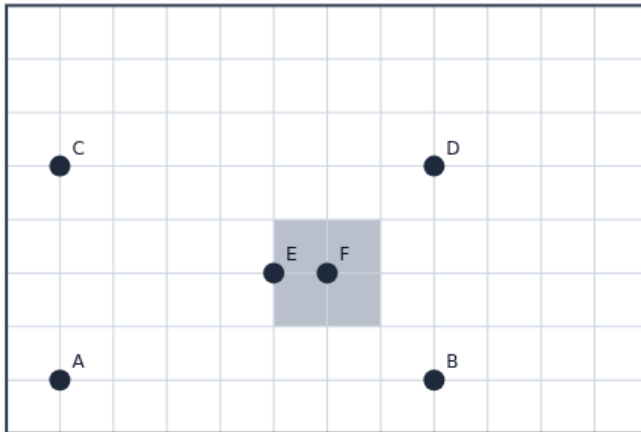
Breadth-first

Around macro

LEE MAZE

Around macro

Step 2 / 5 · detour · module02-01-lee-maze



Explanation

Path (4,1) → (7,1) detours below blockage.

- Avoid blocked cells

len=4

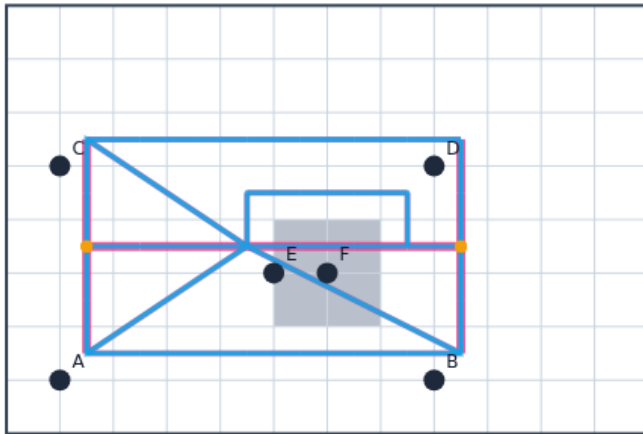
Around macro

Route Lee

LEE MAZE

Route Lee

Step 3 / 5 · lee · module02-01-lee-maze



Explanation

Sequential Lee assigns layers per step.

- Six nets

Finite cost

Route Lee

vs layered

LEE MAZE

vs layered

Step 4 / 5 · compare · module02-01-lee-maze

Explanation

Lee may differ from L overflow totals.

- Compare totals

Lee total≈0

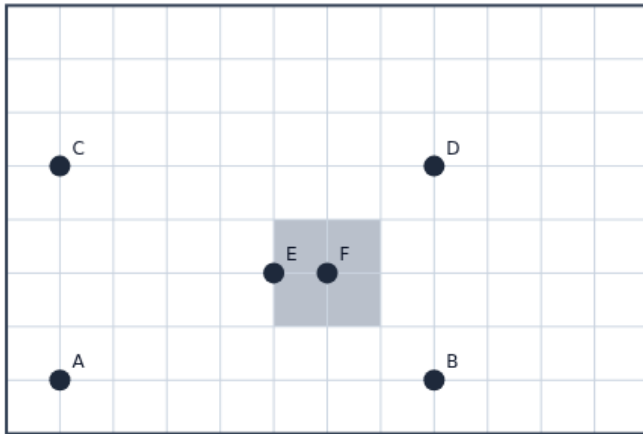
vs layered

When to maze

LEE MAZE

When to maze

Step 5 / 5 · use · module02-01-lee-maze



Explanation

Use maze when pattern routes hit blockages or hot tracks.

- Detailed stage

A* next

When to maze

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Lee maze routing

INTERACTIVE TOOL

Lee maze routing

Breadth-first maze detours around the 2×2 blockage on the routing grid.

Your job: Lee maze routes around blockage. Detour (4,1) → (7,1) length ≈ 4.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Detour length

Pick a challenge and click **Start**. Move cells, route, then **Check**.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓ Route Lee Route layered Route A*

Rip-up detailed

Graph

Reference / metrics

selected: A @ grid (1,1)
routes: (none – click Route)
overflow total: 0.00 · max: 0.00 · count: 0
track capacity: 2 · mode: astar
HPWL: 34.0
detour len: 4

Browser lab starter

learn_routing — 01 lee maze

Implement track

- Implement `lee_maze(start, goal, blocked, nx, ny)`
- Block cells inside five comma two two by two and show the path avoids them
- Match the browser overlay

Pitfalls

- Treating blockages as track usage instead of forbidden cells
- Forgetting BFS needs visited on cells not edges
- Returning a path through a blocked cell because neighbor checks were skipped

Your turn

- Pass Lee maze goldens
- Next: A* routing that respects track capacity